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8. A simple pendulum has a time period T_1. When its point of suspension is moved vertically upwards according as $y = k\,t^2$, where y is vertical distance covered and $k = 1\ ms^{-2}$, its time period becomes T_2. Then, $\frac{T_1^2}{T_2^2}$ is (g = 10 m s^{-2}) (IIT 2005)

a) $\frac{5}{6}$ b) $\frac{11}{10}$

c) $\frac{6}{5}$ d) $\frac{5}{4}$

9. An ideal spring of spring constant k, is suspended from the ceiling of a room and a block of mass M is fastened to its lower end. If the block is released when the spring is un-stretched, then the maximum extension in the spring is (IIT 2002)

a) $4\frac{Mg}{k}$ b) $\frac{Mg}{k}$

c) $2\frac{Mg}{k}$ d) $\frac{Mg}{2k}$

10. A pendulum is hung in a very high building oscillates to and fro motion freely like a simple harmonic oscillator. If the acceleration of the bob is 16 ms^{-2} at a distance of 4 m from the mean position, then the time period is (NEET 2018 model)

a) 2 s b) 1 s

c) 2πs (d) πs

11. A hollow sphere is filled with water. It is hung by a long thread. As the water flows out of a hole at the bottom, the period of oscillation will

a) first increase and then decrease

b) first decrease and then increase

c) increase continuously

d) decrease continuously

12. The damping force on an oscillator is directly proportional to the velocity. The units of the constant of proportionality are (AIPMT 2012)

a) kg m s^{-1} b) kg m s^{-2}

c) kg s^{-1} (d) kg s

13. When a damped harmonic oscillator completes 100 oscillations, its amplitude is reduced to $\frac{1}{3}$ of its initial value. What will be its amplitude when it completes 200 oscillations?.

a) $\frac{1}{5}$ b) $\frac{2}{3}$ c) $\frac{1}{6}$ d) $\frac{1}{9}$

14. Which of the following differential equations represents a damped harmonic oscillator ?

a) $\frac{d^2y}{dt^2}+y=0$ b) $\frac{d^2y}{dt^2}+\gamma\frac{dy}{dt}+y=0$

c) $\frac{d^2y}{dt^2}+k^2y=0$ d) $\frac{dy}{dt}+y=0$

15. If the inertial mass and gravitational mass of the simple pendulum of length l are not equal, then the time period of the simple pendulum is

a) $T=2\pi\sqrt{\frac{m_i l}{m_g g}}$

b) $T=2\pi\sqrt{\frac{m_g l}{m_i g}}$

c) $T=2\pi\frac{m_g}{m_i}\sqrt{\frac{l}{g}}$

d) $T=2\pi\frac{m_i}{m_g}\sqrt{\frac{l}{g}}$

Answers:

1) d	2) c	3) a	4) b
5) b	6) b	7) a	8) c
9) c	10) d	11) a	12) c
13) d	14) b	15) a	